

Chapter 2 Poisson Processes

2.1 Review of Exponential Distribution & Poisson Distribution

Exponential Distribution: Definition & Properties

Exponential distribution with rate λ , $T \sim \exp(\lambda)$:

$$\text{distribution } F(t) = P(T \leq t) = \begin{cases} 1 - e^{-\lambda t}, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

$$\text{or density } f_T(t) = \begin{cases} \lambda e^{-\lambda t}, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

- $ET = 1/\lambda$, $ET^2 = 2/\lambda^2$, $\text{var}(T) = 1/\lambda^2$
- Memoryless: $P(T > s+t | T > s) = P(T > t)$, $s, t > 0$.

- Sums of Exponentials: $\tau_1, \tau_2, \dots$ are ind. $\exp(\lambda)$.

$$T_n = \sum_{k=1}^n \tau_k \sim \text{gamma}(n, \lambda), \text{ i.e.,}$$

$$f_{T_n}(t) = \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!} 1_{t \geq 0}.$$

- Exponential races: $T_i \sim \exp(\lambda_i)$ are ind.,
 $V = \min\{T_1, \dots, T_n\}$, $I = \{i : T_i = V\}$. Then

$$I, V \text{ ind.}; V \sim \exp\left(\sum_{k=1}^n \lambda_k\right), P(I=i) = \frac{\lambda_i}{\sum_{k=1}^n \lambda_k}.$$

($I = i$ with bigger $\lambda_i \nRightarrow V$ is smaller.)

Review of Poisson Distribution

Poisson distribution with parameter λ , $X \sim \text{Poisson}(\lambda)$:

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \dots$$

- $EX(X-1)\cdots(X-k+1) = \lambda^k$, $k \geq 1$;
especially, $EX = \text{var}(X) = \lambda$.
- $X_i \sim \text{Poisson}(\lambda_i)$ independent:
 $X_1 + \cdots + X_k \sim \text{Poisson}(\lambda_1 + \cdots + \lambda_k)$.
- $\text{Binomial}(n, \lambda/n) \xrightarrow{n \rightarrow \infty} \text{Poisson}(\lambda)$.

2.2 Defining the Poisson Processes

Equivalent Definitions of Poisson Processes

Poisson process with rate λ (1st Definition)

$\tau_1, \tau_2, \dots$ i.i.d. $\sim \exp(\lambda)$: time between arrivals.

$T_n = \sum_{k=1}^n \tau_k$: time of the n^{th} arrival; $T_0 = 0$.

$N(s) = \max\{n : T_n \leq s\}$, $s \geq 0$: # arrivals before time s .

Then, $N(s)$ is a Poisson process with rate λ .

- $T_n \sim \text{gamma}(n, \lambda)$, $N(s) \sim \text{Poisson}(\lambda s)$.
- $\{N(t+s) - N(s)\}_{t \geq 0}$: rate λ PP, ind. of $\{N(r)\}_{0 \leq r \leq s}$.
- Independent increments: for $t_0 < t_1 < \dots < t_n$, $N(t_1) - N(t_0), \dots, N(t_n) - N(t_{n-1})$ are independents.

Poisson process with rate λ (2nd Definition)

$N(t)$ is a Poisson process with rate λ , if

- (i) $N(0) = 0$;
- (ii) $N(t)$ has independent increments;
- (iii) $N(t + s) - N(s) = \text{Poisson}(\lambda t)$.

Poisson process with rate λ (3rd Definition)

Counting process $N(t)$ is a PP with rate λ , if

- (i) $N(0) = 0$;
- (ii) $N(t)$ has independent increments;
- (iii) $P(N(t+h) - N(t) = 1) = \lambda h + o(h)$,
 $P(N(t+h) - N(t) \geq 2) = o(h)$.